

ANESS RAHMANI

Backend Developer | FastAPI | PostgreSQL | Express | Redis | Docker

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Backend developer specializing in FastAPI, relational database design, and scalable API architecture. Built and deployed SecureAuth API, a production auth service (JWT, Google OAuth2, bcrypt, Redis-backed sessions, RBAC/ABAC, rate limiting), and a full e-commerce backend implementing caching, async task processing, and containerized deployment. Comes from a strong ML/telecom research background (thesis scored 19.5/20), now focused full-time on backend engineering. Trilingual: Arabic (native), English (C2), French (B2-C1).

TECHNICAL SKILLS

Backend: FastAPI, Python, Node.js, Express.js, RESTful API design, Pydantic, async/await
Databases: PostgreSQL, SQLAlchemy/SQLModel, Alembic migrations, indexing, query optimization, MongoDB
Caching & Messaging: Redis (cache-aside, TTL strategy, rate limiting), Celery (async task queues, retries)
Auth & Security: JWT (access/refresh rotation), OAuth2/OIDC (Google sign-in), bcrypt, RBAC/ABAC, CORS, input validation
DevOps & Testing: Docker, Docker Compose, multi-stage builds, Pytest, GitHub Actions basics
Frontend: React, Next.js, Angular, TypeScript, JavaScript (ES6+), HTML5, CSS3
Tools: Git, GitHub, Postman, Render (deployment)
Languages: Arabic (native), English (C2), French (B2-C1)

PROJECTS

E-Commerce Backend Platform | *FastAPI · PostgreSQL · Redis · Celery · Docker* | 2026

- Designed a fully relational schema (users, products, orders, order items) with SQLAlchemy/SQLModel and Alembic-tracked migrations, including one-to-many and many-to-many relationships and indexed lookup fields
- Integrated SecureAuth API's JWT/OAuth2/RBAC module for registration, Google sign-in, and role-gated endpoints
- Implemented Redis cache-aside pattern on product listing endpoints and Redis-backed rate limiting on auth routes to mitigate brute-force attempts
- Offloaded checkout processing (order confirmation, mock invoice generation) to Celery background workers so API responses return instantly instead of blocking on slow operations
- Containerized the full stack (API, PostgreSQL, Redis, Celery worker) with Docker Compose for one-command local setup; wrote Pytest coverage for authentication flow and core CRUD endpoints
- Deployed live on Render with a Next.js frontend, handling environment configuration and CORS across services

SecureAuth API | *FastAPI · JWT · OAuth2 · Redis* | 2026

- Built a standalone authentication service supporting email/password signup with bcrypt hashing and Google OAuth2/OIDC sign-in
- Implemented JWT access/refresh token flow, Redis-backed session storage, and RBAC/ABAC middleware for route-level permission control
- Added rate limiting, Pydantic request validation, and CORS configuration; Dockerized and deployed as a reusable auth module

AcciSense — Real-Time Accident Detection Platform | *FastAPI · React · MQTT · WebSocket · Supabase* | 2025 – 2026

- Master's thesis project (defended at 19.5/20): benchmarked three video-understanding architectures (R3D-18, SlowFast R50, Video Swin Transformer) for road accident detection and severity estimation, reaching 87.3% detection accuracy on a custom dataset
- Engineered the surrounding production platform: FastAPI backend, React/Vite frontend with real-time WebSocket updates, autonomous MQTT monitoring pipeline, and Supabase logging

Harbor Fish Log — Client Delivery | *React · FastAPI · MongoDB* | 2025

- Delivered a mobile-optimized field data logging app for a PhD researcher, replacing manual spreadsheet workflows; records log in under 30 seconds with instant search, filter, and CSV/JSON export

EXPERIENCE

Web Developer | Freelance Client | 2025 | Remote

- Designed, built, and delivered Harbor Fish Log end-to-end, from requirements gathering through deployment, for a doctoral researcher

Data Annotation Specialist | Freelance (Upwork / LinkedIn) | 2022 – 2024 | Remote

- Completed data cleaning, annotation, and proofreading contracts; standardized a database of thousands of marine species records for a research project

IT Department Head | **University Club** | 2021 – 2023 | Khemis Miliana

- Led technical initiatives and managed digital infrastructure for the organization

EDUCATION

Master's Degree in Telecommunication Systems | **Université Djilali Bounaama** | 2024 – 2026 | Algeria

- Thesis: Automatic Road Accident Detection and Severity Estimation Using Deep Learning Models — 19.5/20

Bachelor's Degree in Electrical Engineering | **Université Djilali Bounaama** | 2020 – 2023 | Algeria